PARKING SYSTEM

OBJECT-ORIENTED PROGRAMMING

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DESCRIPTION OF THE SYSTEM

Purpose

The purpose of this system is to manage parking slots in an organized and efficient way. It helps users reserve parking spaces, park vehicles, track parking duration, and calculate payment based on slot type and time.

Main Features

- Register customer through app/website
- Show all parking slots
- Reserve a parking slot before arrival
- Verify reservation at the gate
- Track parking duration
- Calculate payment
- Detect if a customer parks in a different slot from the reservation
- Support sunny, shaded, and disabled-only slots

LIST OF MAIN CLASSES

01. Customer
02. ParkingSlot
03. SunnySlot
04. ShadedSlot
05. Reservation
06. ParkingSession
07. payment
08. ParkingGateSystem

RESPONSIBILITY OF EACH CLASS

Customer

Represents the user and stores their details.

ParkingSlot

Base class that represents a parking space

SunnySlot

A type of parking slot with a lower hourly rate.

ShadedSlot

A type of parking slot with a higher hourly rate.

Reservation

Links a customer to a slot and checks wrong parking

ParkingSession

Tracks parking time and duration

Payment

Calculates cost based on time and violations

ParkingGateSystem

Controls the whole system (entry, parking, payment)

RELATIONSHIPS BETWEEN CLASSES

Inheritance	Composition	Aggregation
• SunnySlot inherits from ParkingSlot • ShadedSlot inherits from ParkingSlot	• ParkingGateSyste contains slots, reservations, sessions, and payments.	• ParkingGateSystem uses Customer objects, which exist independently

WORK DISTRIBUTION

Noor al rawahi (24-0680) :

Customer Class + Data Handling

Rayyan Alabdul salam (24-0413) :
ParkingSlot (Base Class) + Inheritance (SunnySlot & ShadedSlot)

Shatha AL busaidy (24-0143) :
Reservation Logic + Wrong Slot Detection

Mazayin al busaidi (24-0572) :
ParkingSession (Timer) + Payment Calculation

Basaier Al kharusi (24-0071) :
ParkingGateSystem (Main Controller) + System Integration